

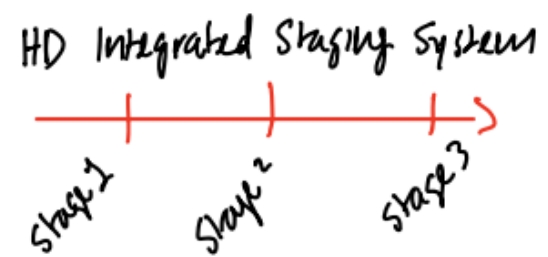
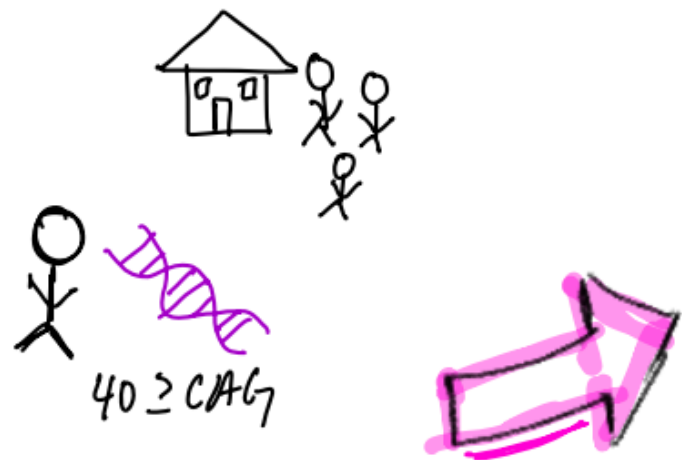
Adjusting for an unmeasured confounder using a right censored event time proxy

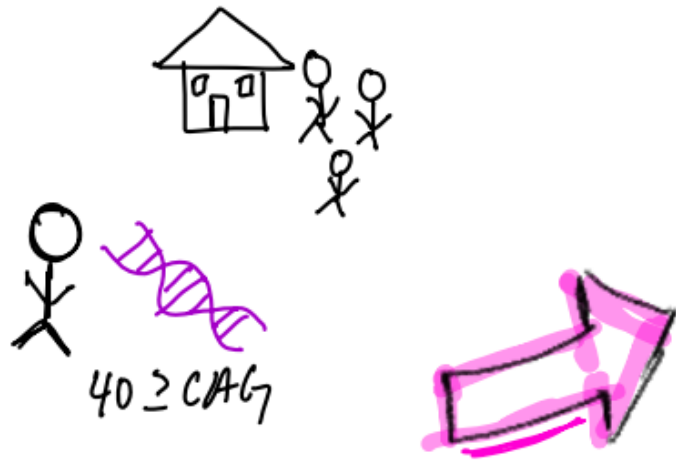
JHU BIOS Retreat | Baltimore, MD
October 2, 2026

Jesus E. Vazquez, PhD | JHU Provost Postdoctoral Fellow

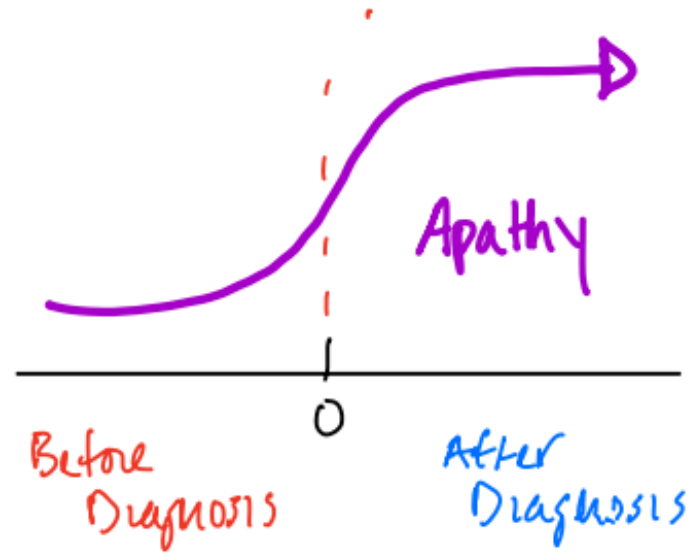
joint work with Grace Ringlein, Jee Bang, Elizabeth A. Stuart, Trang Nguyen







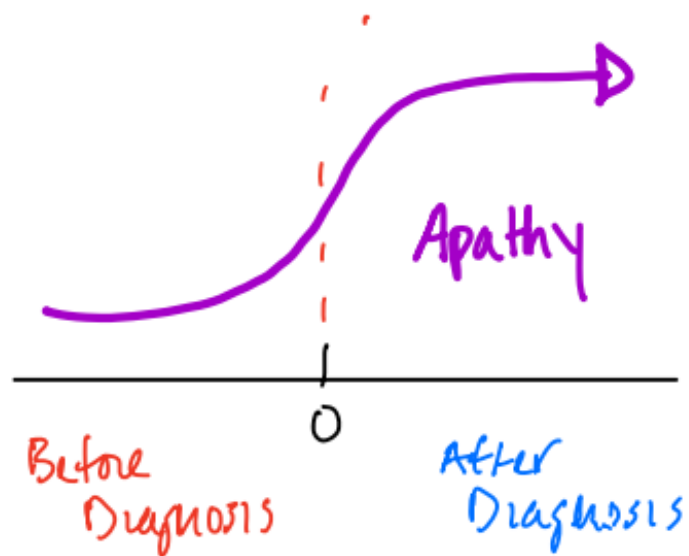
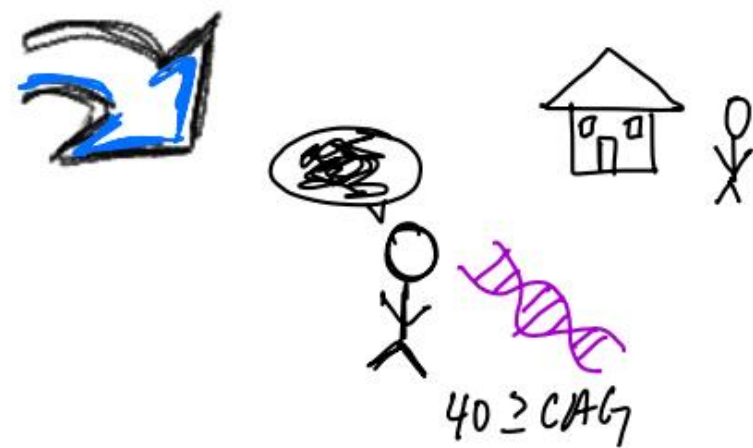
HD Integrated Staging System

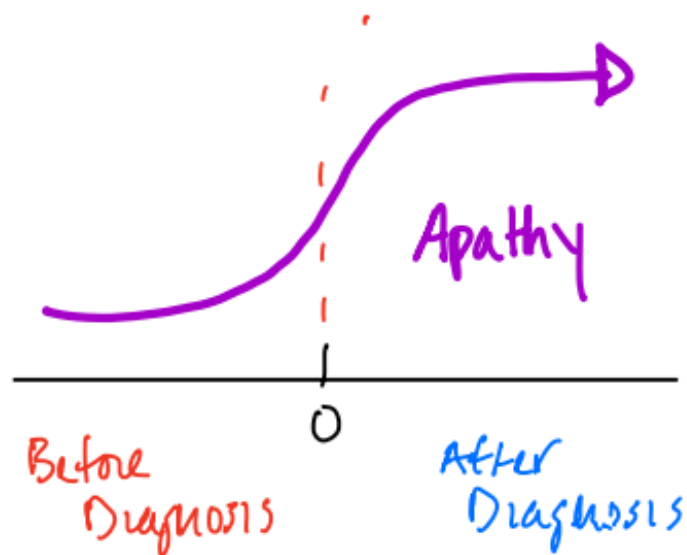
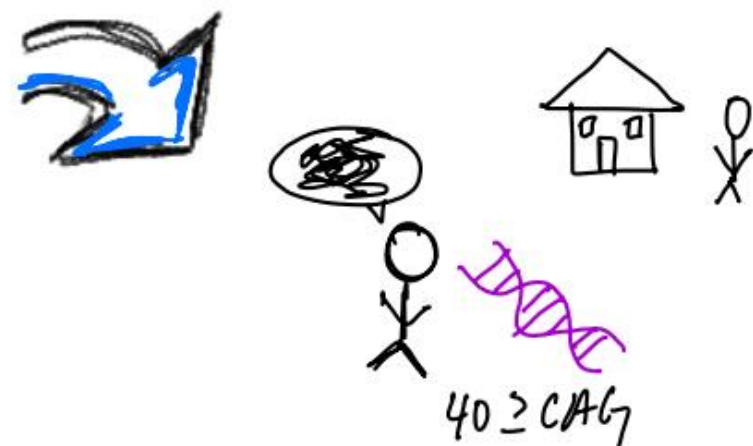
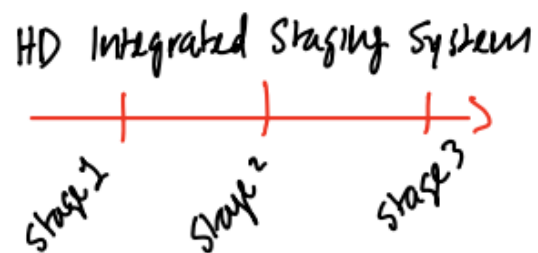
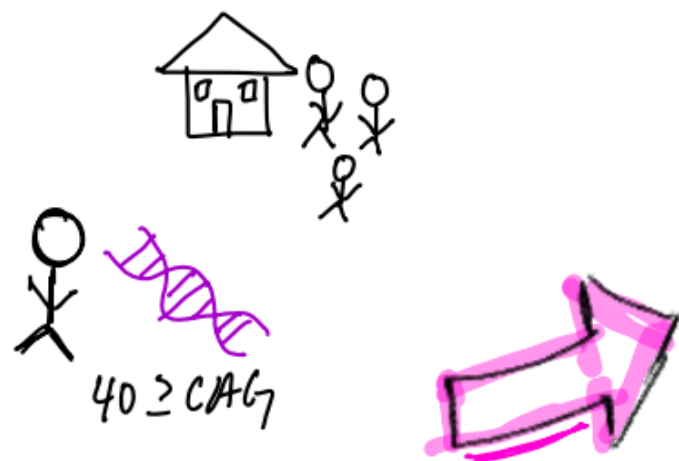




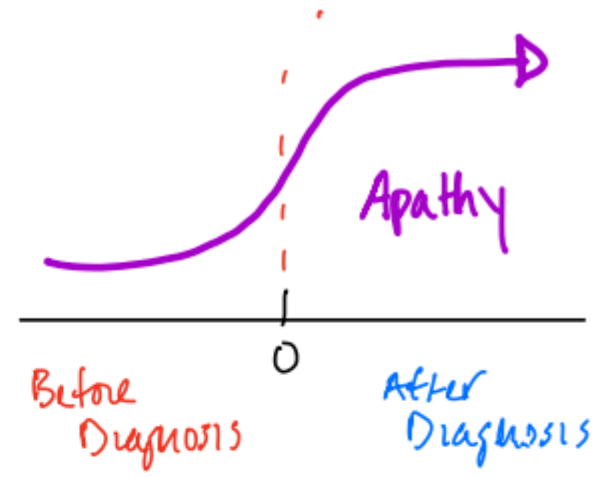
HD Integrated Staging System

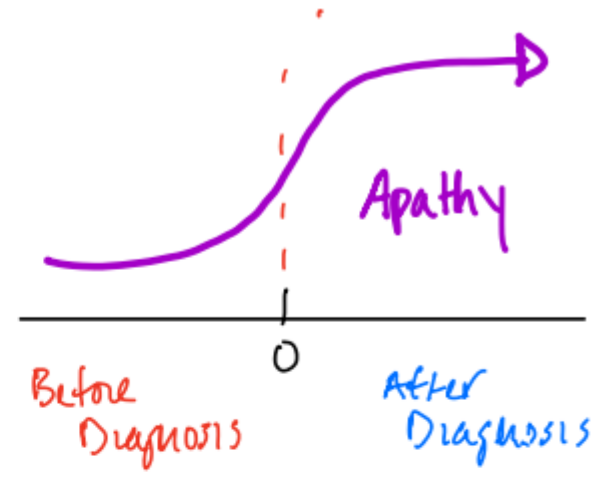
A horizontal red line with three tick marks. Below the first tick mark is the label "stage 1", below the second is "stage 2", and below the third is "stage 3". The line ends with an arrowhead pointing to the right.

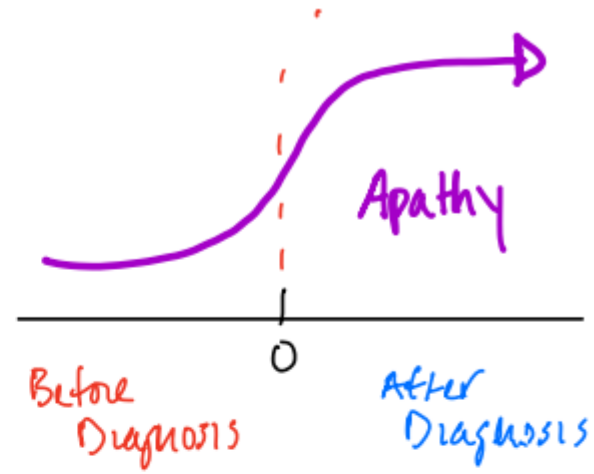




↑ Antidepressant medications







Disease severity (U)



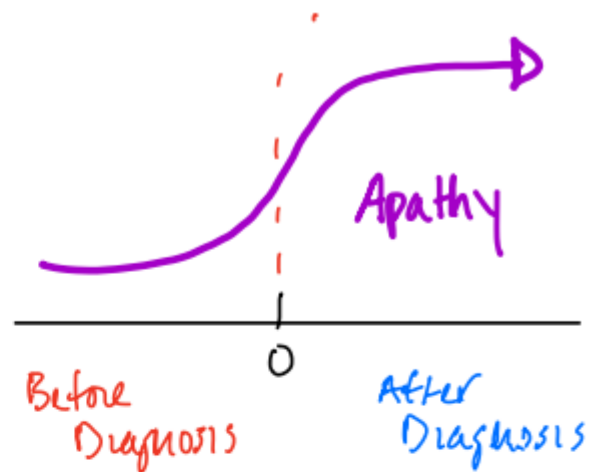
Treatment (A)

(Antidepressants)



Mental health (V)





Disease severity (U)



Treatment (A)

(Antidepressants)



UNMEASURED
CONFOUNDER



Mental
health (Y)

How to adjust for disease severity?



WE CAN USE a proxy that carries similar info as U

$$R = \frac{\text{Age at disease onset}}{\text{Current Age}}$$

How to adjust for disease severity?



WE CAN USE a proxy that carries similar info as U

$$R = \frac{\text{Age at disease onset}}{\text{Current Age}}$$

Example 1: Age at disease onset: 25
Age at treatment initiation: 50

$$\Rightarrow R = \frac{25}{50} = 0.5$$

Example 2: Age at disease onset: 10
Age at treatment initiation: 50

$$\Rightarrow R = \frac{10}{50} = 0.2$$





— STATISTICAL PROBLEM —

A = Treatment (0,1)

Y = Mental Health (1 year)

Z = Age at treatment initiation

X = Age at disease onset

However, instead of observing X , we observe
 (W, Δ) such that $W = \min(X, c)$ $\Delta = I(X \leq c)$



— STATISTICAL PROBLEM —

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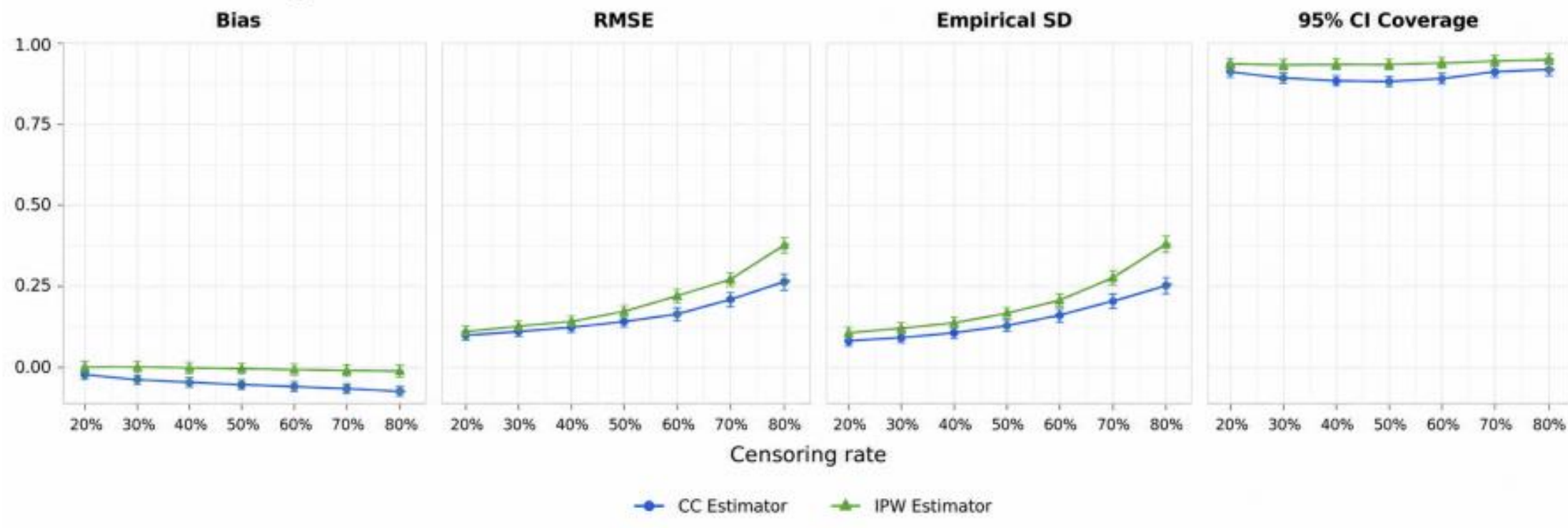
However, instead of observing X , we observe (W, Δ) such that $W = \min(X, c)$ $\Delta = I(X \leq c)$

Important because we need

$$R = \frac{\text{Age at disease onset}}{\text{Current Age}} = \frac{X}{Z}$$

but we have a bounded $\frac{W}{Z}$

After many assumptions:



Thank You!

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